

Satyam Vinayak

Software Developer

CONTACT

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EDUCATION

B.Tech — CSE(AI/ML)

GLA University · 2022–2026
66.20%
Class X - St.Michael's High School 86.60%
May 2019
Class XII — Modern School 73.8%
July 2021

LANGUAGES

- Java · Python · SQL
- JavaScript · HTML · CSS

FRAMEWORKS

- REST API
- React · Flask
- FastAPI
- Next.js

AI / ML

- Scikit-learn · NLP
- Deep Learning · YOLO
- CNN · LLM Integration
- Classification · Clustering
- Transfer Learning · Python Libraries · Data Manipulation
- Data Preprocessing

CORE

- DSA
- OOPS
- CN · DBMS
- SDLC · Error log Analysis
- Software Troubleshooting

TOOLS & INFRA

- Git
- SQL Server
- Sentry · Python Libraries
- PowerBI
- MS Excel

SOFT SKILLS

- Excellent Communication skills
- Problem Solving
- Innovative Thinking
- Adaptability
- Quantitative Aptitude

PROFESSIONAL SUMMARY

Aspiring Software Engineer specializing in Data Science AI/ML with strong foundations in Java, Data Structures & Algorithms, and Object-Oriented Programming. Hands-on experience building deep learning models (LSTM, ANN), NLP pipelines, and RESTful backend systems using Python and Flask. Published researcher in transfer learning-based image classification. Proven ability to design end-to-end solutions — from data ingestion to deployment — and apply OOP principles in real-world projects like a Java-based Banking Management System. Seeking a software development role to contribute both core engineering depth and applied ML expertise.

EXPERIENCE

AI / ML Trainee · GLA University, Mathura Jun – Aug 2024

- Gained hands-on experience in Python, Machine Learning and Deep Learning
- Built classification and clustering models using machine learning techniques
- Post completing the training, created a Project named “**E-mail spam detection model**”, by using Logistic Regression, NLP and Artificial Neural Networks

PROJECTS

AI Code Analyzer · AI Code Maintenance and refactoring Agent Group

Stack: Python, NLP, FASTAPI, FLASK, CI/CD Basics

- Developed an automated code maintenance and refactoring agent using static analysis techniques
- Integrated tools like **Pylint/Flake8** to enforce coding standards and improve overall code quality
- Created a modular pipeline: code input → parsing → issue detection → refactoring → validation.
- Reduced code complexity and improved maintainability by applying **cyclomatic complexity** analysis and optimization techniques
- Designed and developed an interactive frontend interface using Flask, enabling users to upload code files, visualize detected issues, and review suggested refactorings in real time; seamlessly integrated the UI with the backend analysis engine through RESTful APIs, improving usability and end-to-end workflow efficiency.

EduPulse · Student Dropout Prediction System Solo

Stack: Flask · Scikit-learn · OpenCV · Gemini LLM · SQLite · Bootstrap

- Developed a Flask app predicting dropout risk via ML models with automated PDF report generation.
- Integrated Gemini LLM + OpenCV video analysis to produce student engagement Excel summaries
- Implemented Flask-Login for complete session management (login / signup / logout).

Predictive Maintenance System · Predictive Maintenance System using LSTM Solo

Stack: Flask · Scikit-learn · OpenCV · Gemini LLM · SQLite · Bootstrap

- Built a deep learning model using LSTM to predict potential equipment failures and maintenance needs from sensor data.
- Visualized training performance and model predictions using plots and evaluation metrics (accuracy, precision, recall)
- Built an end-to-end pipeline including data ingestion, model training, and prediction output..

RESEARCH & PUBLICATIONS

Refined Horizons: A Comparative Study of Optimized Image Classification Through Transfer Learning

Maheshwari, Rai, Tiwari, Singh & Vinayak · arXiv:2403.10259 · 2024

